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CLINICAL REPORT

An Intensive Design Investigation of Eye Movement Desensitization and Reprocessing of Claustrophobia

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Abstract — Two claustrophobic subjects were treated with eye movement desensitization and reprocessing (EMDR), a new treatment for induced anxiety disorders. Both subjects were interviewed to construct detailed images of fear-related events before treatment. The treatment followed a within-series phase-change design to examine the effects of eye movement added to the general treatment protocol. Both subjects' verbal reports of fear changed substantially when eye movements were added to the general treatment protocol. It was concluded that the addition of eye movement was necessary to reduce the aversiveness of some phobic imagery. No change in heart rate was observed for either subject. These results are discussed within the framework of the growing number of EMDR outcome studies.

Since the first published account (Shapiro, 1989), the promotion of eye movement desensitization has prompted a large number of clinical case reports describing the effects of treatment. However, three methodological reviews (Acierno, Hersen, Van Hasselt, Tremont, & Mueser, 1994; Herbert & Mueser, 1992; Lohr, Kleinknecht, Conley, Dal Cerro, Schmidt, & Sonntag, 1992) concluded that empirical support for its efficacy is meager.

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Following the critiques, several controlled experiments with improved methodologies have been published. Boudewyns, Stwertka, Hyer, Albrecht, and Sperr (1993) randomly assigned veterans' hospital patients with PTSD to one of three groups: EMDR, exposure control, or milieu-only control group. Results indicated that posttreatment subjective units of discomfort (SUD) ratings were lowest for the EMDR group. Therapist ratings of treatment responders versus nonresponders also favored the EMDR group. However, standardized self-report measures showed no differential effect of treatment, and no form of treatment appeared to affect physiological measures of anxiety.

In a controlled treatment outcome study, Jensen (1994) randomly assigned combat veterans with PTSD to either an EMDR condition or a milieu control group. As with Boudewyns et al. (1992), analysis of the SUD ratings revealed lower scores in the EMDR group. However, the results also showed there was no difference between the two groups on standardized measures of posttraumatic stress disorder (PTSD) or on the attainment of individualized treatment goals.

Two additional studies have compared the effect of EMDR with control conditions manipulating eye movement content (Renfrey & Spates, 1994) and with alternative treatments (Vaughan et al., 1994). Renfrey and Spates (1994) showed reductions in verbal report and standardized measures of symptoms with EMDR, but the reductions were no greater than those in the control conditions. Vaughan et al. (1994) compared EMDR to imagery habituation training (Vaughan & Tarrier, 1992) and applied muscle relaxation training (Öst, 1987). Impact of event-intrusion scores were reduced at posttreatment only in the EMDR group, but were no different from the other conditions at a 3-month follow up. Comparisons between treatments on PTSD symptoms and depression showed reliable reductions from pretreatment to posttreatment but no differential effect of treatments. Despite the absence of the necessary treatment by measurement interaction, post hoc multiple *t*-tests showed that EMDR resulted in fewer flashbacks and nightmares and avoidance symptoms. Because the *t*-tests increase the risk of Type I error, however, the effects of EMDR must be interpreted with caution. Moreover, the apparent improvement in symptoms could be the result of nonspecific effects of treatment for which no controls were implemented.

Montgomery and Ayllon (1994a) have reported the first single-subject experiment investigating the effect of EMDR on PTSD. A multiple baseline across images was used to treat separate traumatic incidents of an automobile accident and an armed robbery. Visual inspection of the within-series (A-B) manipulation on each image revealed parallel reductions in SUD ratings, heart rate, and systolic blood pressure during the EMDR phase for each image. Comparisons between images revealed that the effective treatment of the first image did not generalize to the second image. A similar study using a multiple baseline design (Acierno, Tremont, Last, & Montgomery, 1994) failed to show any change in behavioral, physiologic, or cognitive assessment domains.

The second study (Montgomery & Ayllon, 1994b) used a multiple baseline across six individuals diagnosed with PTSD. Within-series treatment phase

changes were designed to assess the effect of adding components of the EMDR procedure. In the baseline (A) phase, subjects were exposed to traumatic imagery with the accompanying negative cognition, SUD ratings were obtained, and positive cognitions were constructed. In the first intervention (B) phase, the specific EMDR orientation instructions were added to the baseline content. In the second intervention phase (B+C), eye movements were added to provide the complete EMDR procedure. The last phase was a follow-up in which assessments were made without any specific treatment content.

In contrast to the findings of Montgomery and Ayllon (1994a), Montgomery and Ayllon (1994b) found no statistically significant changes with physiological measures. A *t*-test showed a statistically significant effect for repeated assessment on SUD ratings, but no significant effect for between-treatment phases. Thus, it appears that reduced SUD ratings were a function of time and/or repeated assessment. Visual inspection of SUD ratings across additive treatment phases showed no change in one subject, equivocal change in three subjects, and distinct change in only two subjects.

Given these findings, the extension and application of EMDR to other disorders should proceed with caution. Nonetheless, EMDR has been claimed to be effective for an expanding range of disorders that include multiple and diffuse anxieties (Marquis, 1991; McCann, 1992; Pellicer, 1993; Wernik, 1993), depression (Marquis, 1991; Puk, 1991), and many other clinical presentations. The application of EMDR to disorders that share etiologic and symptomatic features of PTSD, such as traumatically acquired phobias, would represent a more cautious clinical extension. For example, Kleinknecht (1993) has reported the case of a 21-year-old woman with a blood/injection phobia that was acquired during a painful procedure during childhood. EMDR treatment was administered over four treatment sessions and resulted in substantial reductions of SUD ratings, self-reported fear, blood pressure, and heart rate both within and across sessions. Following treatment, the subject was able to seek and obtain a flu shot and blood draw as an indication of treatment effectiveness. These positive changes persisted at one-, 14-, and 24-week follow-up assessments.

Sanderson and Carpenter (1992) conducted a group crossover design experiment with 58 phobic individuals. All subjects were administered EMDR and a control procedure that was identical to EMDR except that eye movement was deleted. Analysis of feared imagery ratings showed that both EMDR and the control procedure resulted in lower ratings after treatment, with no reliable difference between the two conditions. These results indicate that the some aspect of the EMDR procedure may be beneficial in the treatment of phobia.

Following our review of the effectiveness of EMDR (Lohr et al., 1992), we recommended the application of intensive design research as an experimental alternative to uncontrolled case studies and as precursors to more extensive group-design studies. The use of single-subject experimental designs could effectively test EMDR treatment outcome with a variety of specific diagnostic conditions. These investigations could employ simple phase-change (A-B-A)

designs to compare the effect of an imagery-only baseline phase (A) to an EMDR treatment phase (B), with a return to baseline (A). The comparisons may be replicated across a number of traumatic images to determine the degree of treatment generalization. In this time-series experiment, we describe the application of EMDR to traumatically induced claustrophobia.

EXPERIMENT 1: METHOD

Subject 1

Subject 1 was a 68-year-old Caucasian male with a 49-year clinical history. He met *DSM-IV* criteria for specific phobia, situational type (claustrophobia). In addition to his fear of enclosed situations, he also feared experiencing the resultant panic attacks themselves. The claustrophobic situations addressed here included aircraft travel, elevators, and crowded rooms and recalled imagery of the original precipitating incident of being confined below the waterline of an aircraft carrier. He had never sought psychological services for his psychological difficulties. During the prior 18 months he had been on a daily regimen of 100 mg of Sertraline HCl (Zoloft) prescribed by his family physician.

Materials and Procedure

Subject 1 was recruited from the community through announcement of services available. Pretreatment assessment consisted of an interview to obtain informed consent, clinical history, and administration of the SCL-90-R (Derogatis, 1993). The phobic memories and images that would serve as the content for the EMDR procedure were reconstructed during the second interview. Thematic content was differentiated, and preliminary drafts of imaginal scenes were written. The subject was instructed, as a homework assignment, to refine the content and reconstruct additional scenarios, should he recall them. In the third session, scenes were refined to include a specific "picture" that would summarize the significance of the fear. It was to this "picture" that a "negative cognition" was assigned. A positive cognition was also generated, to which a validity of cognition (VoC) rating as assigned during treatment.

Subject 1 produced three scenarios, the first of which (Image 1) was one that had been accompanied by a panic attack. It had occurred 20 years earlier at a business meeting attended by 25 people in a small, smoke-filled conference room. A second scene (Image 2) recalled an incident that took place 10 years earlier, in which he experienced panic during an elevator ride in his place of business. Image 3 occurred when he was a 19-year-old naval recruit first assigned to quarters below the water line on an aircraft carrier. The salient features of the scene were the extremely close living quarters and the locking of water-tight hatches as they moved below deck. He considered this to be the etiological event of his disorder. These several clinical concerns impaired his current functioning enough that he avoided activities that required

reported that it was one of his best performances as he was able to concentrate on artistic expression rather than ruminating about merely "getting through it".

The 6-month posttreatment telephone contact revealed that he had voluntarily attended two large concerts and was seated in the middle of the audience on both occasions. He had experienced a maximum SUD rating of 5 on one occasion and 15 on the other. Previously, he would have been unable to remain seated for the duration of the performance. He had not travelled on a commercial airline and had not obtained the CT scan. He reported feeling generally more calm since the completion of treatment and indicated he was sleeping better.

DISCUSSION

Methodological reviews of EMDR (Herbert & Mueser, 1992; Lohr et al., 1992) have called for the use of single-subject experimental designs for the initial investigations of the technique for various clinical problems. This study is the first to apply time-series methods to the investigation of EMDR with simple phobia. It contains methodological and clinical improvements over time-series investigations of the EMDR treatment of posttraumatic stress disorder. Montgomery and Ayllon (1994a, 1994b) plotted the mean of 20 exposures per treatment session. Because our procedure allows the separate plotting of each data point, it provides for the direct visual comparison of every treatment trial to every adjacent trial without recourse to statistical averaging. In addition, the clinical procedure is the same as that described by Shapiro (1992). The procedure reported by Montgomery and Ayllon (1994a, 1994b) involves many more imagery exposures and many more treatment sessions than does Shapiro's (1992) procedure. We have, therefore, provided a more direct graphic representation of data and a more accurate application of the clinical technique.

The data reveal no consistent effects of any aspect of treatment upon heart rate. This finding is consistent with that of Montgomery and Ayllon (1994b) and Boudewyns et al. (1992) who investigated PTSD. This finding fails to replicate the findings of Montgomery and Ayllon (1994a) who found reductions in heart rate across two images in a single individual with PTSD. It is difficult to reconcile these findings on methodological grounds, as the means and timing of measurement were virtually identical to those used by Montgomery (Montgomery, personal communication, July 7, 1994). It is possible that the difference may be due to the larger number of imagery exposures and treatment sessions used by Montgomery and Ayllon (1994a), but that is unlikely because the same protocol was used in Montgomery and Ayllon (1994b). In any case, it appears that cardiovascular change is not a reliable effect of EMDR.

SUD ratings appear to be responsive to phase-changes in treatment content. With Subject 1, mere repetition was followed by reductions, but these were neither large nor stable. The EMDR analogue procedure appeared to have a substantial influence on two contemporary images, but not on the etiological image. Only when the eye movement was added was there a substantial reduction of SUD ratings. SUD ratings remained low with mere repetition, indicating

was also contacted 6 months following treatment to obtain anecdotal information about current phobic symptoms.

RESULTS

Pretreatment SCL-90-R scores showed that the highest score was Phobia (*T*-score = 59). All other scores were less than a *T*-score of 50. The posttreatment Phobia score was 63, indicating no reliable change.

A number of statistical analyses were considered for the time-series data but were rejected because of design or data constraints.[1] Thus, visual inspection remained as the most meaningful means of evaluating within-series phase-change comparisons. The results for Subject 1 are presented in Figs. 1 and 2. The data are presented for visual inspection with SUD ratings presented on the left ordinate and heart rate presented on the right ordinate. The data reveal no change in heart rate between any two adjacent phases. The SUD ratings suggest that mere repetition (A) of Image 1 and Image 2 was sufficient to show consistent reduction in SUD ratings. Image 3 did not seem to respond to mere repetition. The re-presentation of Image 1 on the second treatment day showed a return to baseline levels, but it was reduced to low levels using the analogue treatment procedure (B). The application of the analogue procedure to the previously unresponsive image (3) also resulted in a relatively rapid and strong reduction in SUD ratings. Because of this clinical response to Images 1 and 2, it was judged by the therapists that the application of the EMDR to these two images would be superfluous. Thus, it appears that the EMDR analogue may have therapeutic effects of its own. Interestingly, however, the analogue procedure did not seem to have any effect upon Image 2. This was the image of the etiological event, and Subject 1 noted that "he could just not get over it like the first two." As a result, it was decided to implement the EMDR procedure (B+C) phase on the next treatment day (Fig. 2). Within nine imagery exposures, the SUD ratings declined to zero and remained for two more saccades. Reexposure of the three images using the baseline procedure produced SUD ratings of very low intensity.

We have also plotted the heart rate and SUD ratings of the etiological ship image (4) in Fig. 3. We should note that this graphic representation is somewhat misleading because no phase is adjacent in real time. Nonetheless, the pattern of data is intriguing. It appears that baseline (A) and analogue (B) procedures reduced SUD ratings, but plateaued at relatively high levels. When the EMDR procedure (B+C) was next applied, ratings dropped quickly and substantially and did not recur at pre-EMDR levels in the baseline phase.

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[1] *T*-test comparisons between phases were rejected because serial dependency was detected for both subjects during one of the B and B+C treatment phases. The split-middle technique of Kazdin (1982) was rejected because there were more observations taken during the EMDR phase, which would have increased the likelihood of a Type I error. The *C*-statistic (Tryon, 1982) was rejected because it is most appropriate for detecting abrupt changes in level and not changes in slope between adjacent phases.

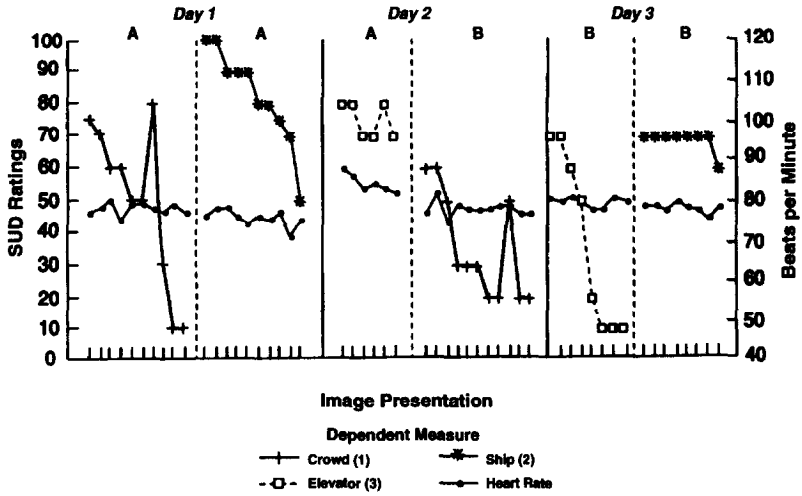

FIG. 1. SUBJECT 1 SUD RATINGS AND HEART RATE DURING BASELINE AND EMD ANALOGUE.

The 6-month posttreatment telephone contact revealed that he had voluntarily used an elevator on the one occasion that presented itself and experienced a maximum SUD rating of 20. He had also participated in a number of small social gatherings, with the same SUD rating. He reported no opportunity to travel by commercial airline or participate in other activities that would have the potential for a fear reaction.

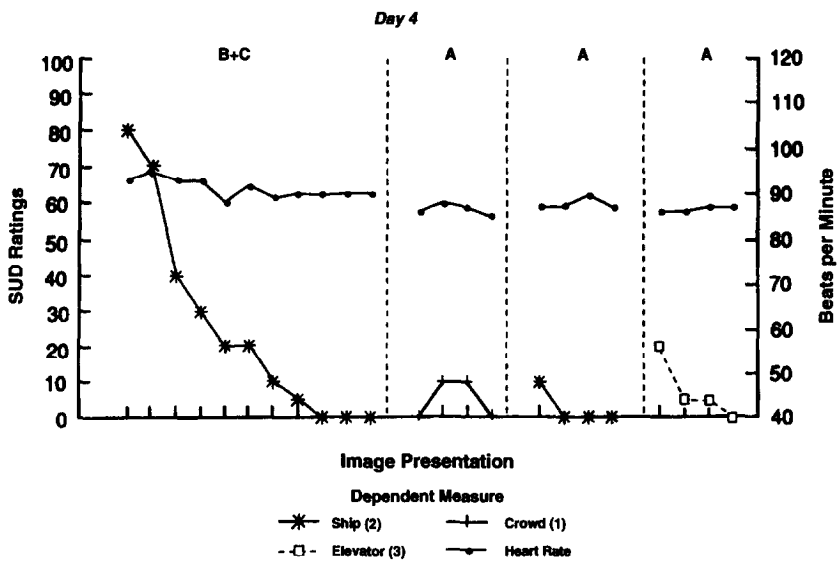

FIG. 2. SUBJECT 1 SUD RATINGS AND HEART RATE DURING EMDR AND BASELINE.

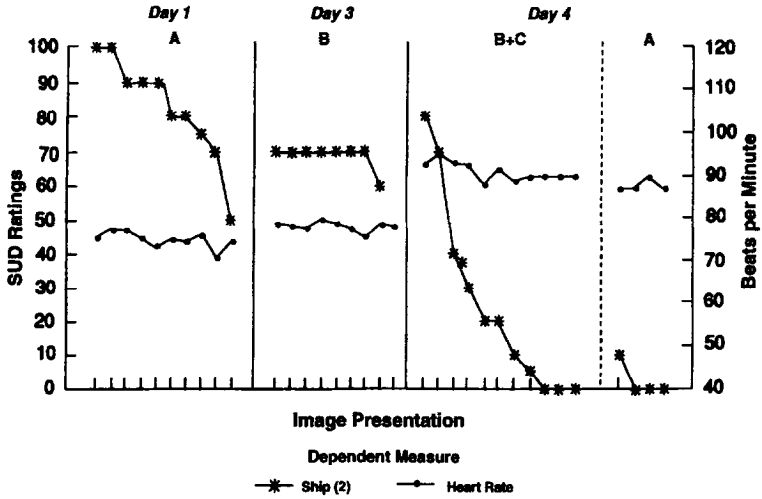

FIG. 3. SUBJECT 1 SUD RATINGS AND HEART RATE DURING BASELINE, EMDR ANALOGUE, AND EMDR OF SHIP IMAGE (2).

EXPERIMENT 2: METHOD

Subject 2

Subject 2 was a 53-year-old Caucasian male with a 33-year clinical history of specific phobia, situational type (claustrophobia), who met *DSM-IV* criteria (American Psychiatric Association, 1994). The claustrophobic content included previous aircraft travel, a recent CT-scan procedure, and the original precipitating incident in which he had been detained in an underground tunnel by Soviet troops in East Berlin. Subject 2 was also a concert pianist who had developed severe performance anxiety that he felt was thematically more akin to confinement fear than to social phobia. A common theme of the phobic content related to physical circumstances where avenues of escape were limited or unavailable or were too costly in terms of social consequences. Similar to Subject 1, fear of panic was a salient component of the clinical presentation.

Subject 2 had availed himself of insight-oriented psychotherapy numerous times since the original event, with little satisfaction. He had never participated in therapist-directed anxiety-management procedures such as systematic desensitization or relaxation training. He had, however, participated in a commercial self-help course for fear of flying, which included a "final exam" flight aboard a commercial aircraft. He found the flight to be a near-traumatic event despite extensive classroom preparation. Three months prior to initiation of the study he had obtained EMDR for the fear of flying from a mental health practitioner who had been recently trained in the procedure. After constructing an image of an anticipated airline flight to attend his daughter's wedding, the modified pro-

tocol was loosely applied with little effect, as he was unable to complete the flight when the time arrived. He responded to the public service announcement of a university-sponsored research program for phobic disorders.

Subject 2 produced four scenarios, the first of which was the precipitating incident at age 20 (Image 1). He was travelling in Europe in the early 1960s and had entered East Berlin via an unauthorized train route to be used only by German workers. Having been discovered by border authorities, he was incarcerated below ground guarded by Soviet troops with automatic weapons. It was during this time that he suffered a panic reaction. He considered this to be the etiological event of his disorder. The other three scenes contained contemporary content. Image 2 recalled a CT-scan procedure to diagnose a spinal disorder. Image 3 recalled a 1 1/2 hr night flight on a commercial aircraft that followed the earlier self-help course. Image 4 recalled a demanding solo concert piano performance. Concerns with fear and panic substantially interfered with the quality of life. He was currently experiencing back pain but was unable to complete the CT scan for fear of a panic reaction. In a similar vein, he was faced with frequent requests to perform at concerts, many of which he declined. Those that he felt compelled to accept were accompanied by intense dread and resulted in performances that he considered unsatisfactory.

Materials and Procedure

The assessment and general treatment procedure was the same as that for Subject 1. In planning the design of treatment for Subject 2, it was decided to implement the experimental protocol first upon the image that the subject considered to be etiological. The sequence of the remaining images was ordered so that the flying fear treated earlier was bracketed by two nontreated fears. This sequence would allow for the investigation of two untreated fears before a reiteration of the flying fear treatment and would allow for the investigation of an untreated fear after the reiteration.

It was decided to implement intervention in a way to permit more thorough investigation of the eye movement component of the procedure by separating the analogue procedure from eye movements for each of the four images. Inclusion of a baseline phase for each image would have increased number of phases by 50% and was judged by the subject and the investigators to be an excessive procedure despite its methodological desirability. Thus, the imagery only (A) phase was dispensed with, and treatment began with the EMDR analogue (B) phase. In the following EMDR phase, the eye movement saccades were added to the procedure (B+C). The EMDR procedure was followed by a probe, with the next image scheduled for EMDR treatment using the analogue (B) procedure. The fourth phase was a return to the analogue procedure using the image that had been subjected to the full treatment in the second phase. The two posttreatment baselines provided for the assessment of forward generalization and for the return of fear. The design is schematized thus: B-B+C-B-B and

was replicated across the four successive images on four successive treatment days. On the fourth day, the assessment probe used was Image 1, the first to undergo the entire sequence of procedures. The treatment days were scheduled approximately one week apart. This design also allowed for the treatment of different images on separate days and for the direct comparisons of adjacent phases. The changes in the design are improvements over the design for Subject 1 because they eliminate the Treatment Day × Treatment Procedure confound, which was inherent in the prior protocol. Posttreatment questionnaire assessment and anecdotal inquiry were the same as with Subject 1.

RESULTS

The Symptom Checklist-90-R showed the highest scale was Phobia (*T*-score = 61). The next highest scale was Interpersonal Sensitivity with a *T*-score of 53. The posttreatment Phobia score was 54. The results for Subject 2 on Images 1, 2, 3, and 4 are shown in Figs. 4 through 7, respectively. As with Subject 1, there appears to be no effect of any within-series phase change upon heart rate. SUD ratings appear to show phase-change differences, but the nature of the changes depends on the image treated. The EMDR analogue (B) applied to Image 1 appears to have induced an increase in subjective discomfort. However, the application of the EMDR procedure was accompanied by an abrupt change in level and a downward slope. A probe with a new image revealed no generalization, and a return to the analogue procedure showed no return of fear.

The replication of the experimental protocol upon Image 2 (Fig. 5) yielded a similar pattern of data. The analogue procedure appeared to increase SUD ratings, and the EMDR procedure led to an abrupt lowering of ratings. The slope of the

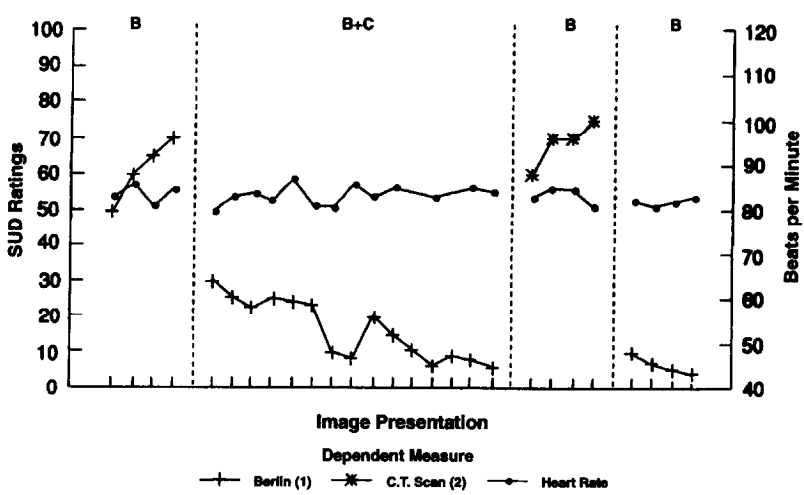

FIG. 4. SUBJECT 2 SUD RATINGS AND HEART RATE DURING EMDR ANALOGUE AND EMDR IN TREATMENT SESSION 1.

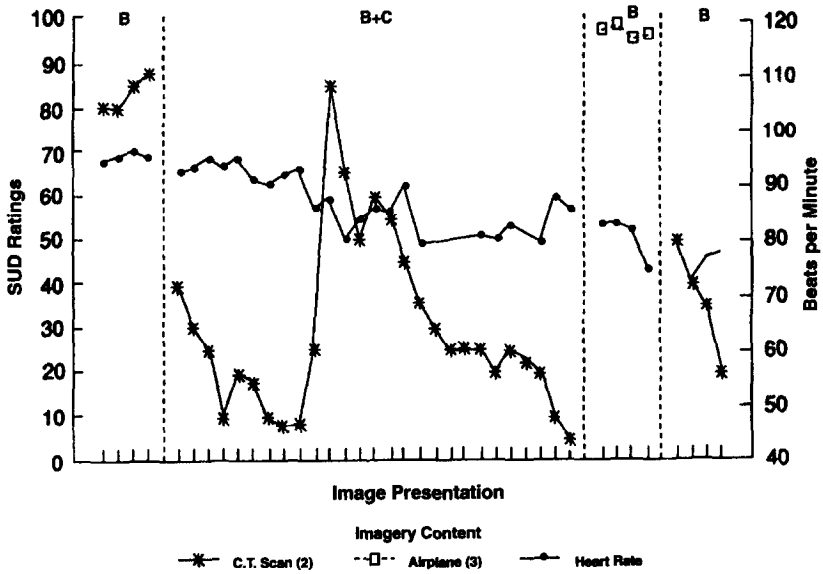

FIG. 5. SUBJECT 2 SUD RATINGS AND HEART RATE DURING EMDR ANALOGUE AND EMDR IN TREATMENT SESSION 2.

SUD ratings was more variable, however. After 10 saccades, SUD ratings reached a peak of pretreatment levels as the image content changed. As EMDR was continued, a gradual reduction of ratings to zero was obtained. The curvilinear process is apparently common in clinical experience with EMDR (Shapiro, 1992). The application of the analogue procedure to the probe image showed no forward generalization of the effect of EMDR, but there appeared to be a moderate return of fear, which decreased with repeated exposure. Replication of the protocol upon Image 3 (Fig. 6) showed a pattern that was very similar to Image 1. The first analogue application had no effect, and the implementation of EMDR was followed by a gradual reduction in SUD ratings. The only real difference appeared in the second analogue phase with the next image. It appeared that the second analogue procedure had an effect of reducing SUD ratings of an image not yet subjected to EMDR. The pattern of data for Image 4 (Fig. 7) is similar to that of Image 3. The analogue procedure was ineffective, but the EMDR phase revealed a gradual reduction in SUD ratings. It should be noted, however, that the application of the analogue procedure to Image 1 showed a recurrence of high SUD ratings. Thus it appears there may have been a return of fear after a period of approximately 4 weeks. The return of fear measured by SUD ratings following EMDR may be idiosyncratic to this individual, as it is a little observered phenomenon in the few controlled studies that have provided follow-up assessment (Renfrey & Spates, 1994; Wilson, Becker, & Tinker, in press).

Behavioral confirmation of the self-reported SUDS reduction was obtained when Subject 2 telephoned us 3 weeks after the last day of treatment on the

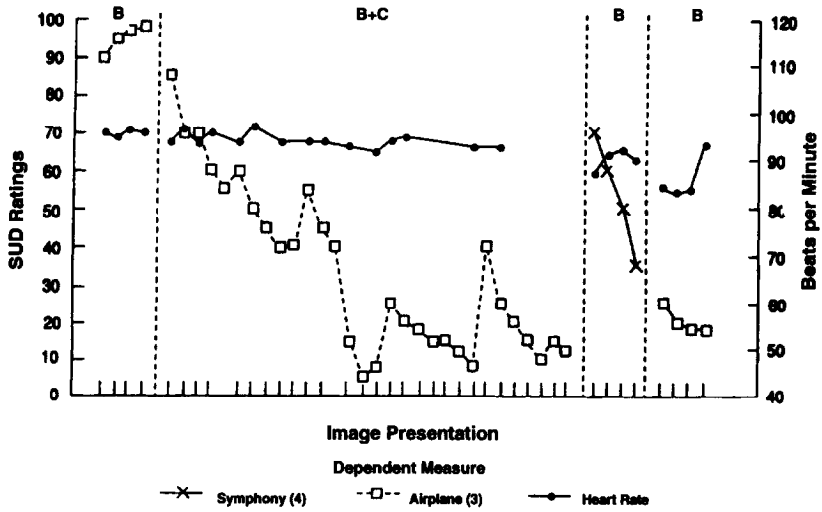

FIG. 6. SUBJECT 2 SUD RATINGS AND HEART RATE DURING EMDR ANALOGUE AND EMDR IN TREATMENT SESSION 3.

concert performance image to inform us of a performance in which he had participated. He reported virtually no anticipatory anxiety leading up to the performance and a maximal SUD rating of 50 during the performance. Previous to this time, he would have experienced SUD levels of 95 to 100. He also

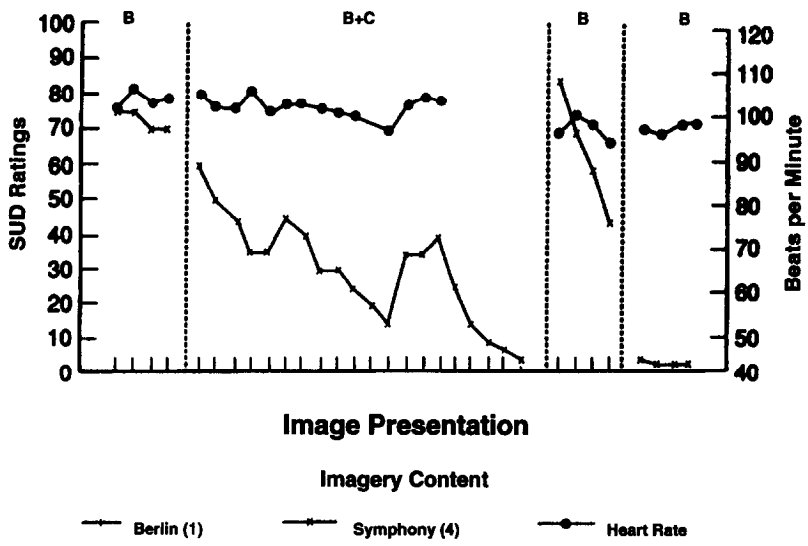

FIG. 7. SUBJECT 2 SUD RATINGS AND HEART RATE DURING EMDR ANALOGUE AND EMDR IN TREATMENT SESSION 4.

reported that it was one of his best performances as he was able to concentrate on artistic expression rather than ruminating about merely "getting through it".

The 6-month posttreatment telephone contact revealed that he had voluntarily attended two large concerts and was seated in the middle of the audience on both occasions. He had experienced a maximum SUD rating of 5 on one occasion and 15 on the other. Previously, he would have been unable to remain seated for the duration of the performance. He had not travelled on a commercial airline and had not obtained the CT scan. He reported feeling generally more calm since the completion of treatment and indicated he was sleeping better.

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SUD ratings appear to be responsive to phase-changes in treatment content. With Subject 1, mere repetition was followed by reductions, but these were neither large nor stable. The EMDR analogue procedure appeared to have a substantial influence on two contemporary images, but not on the etiological image. Only when the eye movement was added was there a substantial reduction of SUD ratings. SUD ratings remained low with mere repetition, indicating

that fear did not return following some or all of the EMDR procedure. Subject 2 showed that the EMDR analogue procedure was ineffective with all images. Only when the eye movement procedure was added did SUD ratings decline. Interestingly, the effect of eye movement seemed to be specific to the image treated, as there was no generalization to untreated images, similar to previous findings with a blood-and-needle phobic (Kleinknecht, 1993). The data also revealed that there is no return of fear immediately following treatment, though there may be some return weeks later. In any case, the analogue procedure seems effective in returning SUD ratings to low levels. In summary, it appears that some aspect of the EMDR procedure facilitates the reduction of verbal reports of phobic emotions and may affect anxiety-related behavioral changes as with Subject 2.

While we did not plan to assess behavioral indices of change, Subject 2 spontaneously reported participating in a musical concert. The performance was analogous (if not identical) to the image that was most recently subjected to EMDR treatment. This was in contrast to previously reported episodes of anxiety and avoidance when asked to present solo concert performances. He reported substantial improvement in both indices of anxiety (SUD ratings) and in the technical aspects of the artistic performance.

A variety of methodological considerations, however, should qualify even this tentative conclusion. While SUD ratings may have psychometric validity, it is also the case that they appear to be the only measures that change when EMDR is applied. The fact that they also change in response to control procedures (Boudewyns et al., 1992; Sanderson & Carpenter, 1992) suggests that they are vulnerable to nonspecific effects such as demand and expectation or that imaginal exposure is sufficient to alter these measures. The investigation of nonspecific effects will require the manipulation of these variables in a group design experiment analogous to that described by Borkovec, Kaloupek, and Slama (1975).

The extension of single-subject methodology to other phobias will help clarify the effects and generality of EMDR. Medical fears are those to which EMDR might be most directly extended because of their frequent traumatic acquisition (Kleinknecht, 1994). A review of the etiology of phobia suggests that a large proportion of phobic individuals can report an incident wherein pain or emotional distress first served to precipitate or greatly exaggerate the pattern of phobic response (Kleinknecht & Lenz, 1989). Öst and his colleagues (Öst, 1989; Öst, Hellstrom, & Kaver, 1992) have developed an in vivo reexposure protocol that effectively treats 80% of needle phobics. However, the procedure may not be practical in most psychological treatment settings because it requires therapist-administered injections or blood draws. A time-efficient treatment that does not require the application of invasive procedures would be more practical for the nonmedical practitioner, and it appears that EMDR may be one such procedure. We are currently conducting a series of single-subject experiments on persons with medical fears manipulating the content of treatment with additive and subtractive phase-change strategies.

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